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MAJOR PROJECT REGISTRATION FORM

Date:

Project Category:	☐ Research oriented ☐ Application oriented ☐ Product oriented ☐ Other (please specify) Specify if others:	Batch ID:	A08

Title of the Project: Multi-Agent Orchestrator for GitHub Issue Resolution

Abstract

Modern software development projects generate a large number of GitHub issues related to software bugs, feature requests, code refactoring, and performance improvements. Resolving these issues requires developers to analyze complex code repositories, identify the affected components, implement appropriate code modifications, validate the changes through testing, and prepare pull requests, making software maintenance a time-consuming and resource-intensive process. This challenge is primarily faced by software developers, open-source contributors, and development teams maintaining medium- and large-scale GitHub repositories. Existing AI coding assistants such as GitHub Copilot and Cursor significantly improve code generation but primarily rely on a single Large Language Model (LLM) and provide limited support for structured, repository-level software engineering workflows. Recent research, particularly the MAGIS framework, has demonstrated that coordinating multiple specialized AI agents can improve GitHub issue resolution through collaborative task execution. Although the MAGIS framework establishes an effective foundation for multi-agent software engineering, its implementation provides limited support for modular workflow orchestration, efficient repository context retrieval, and flexible deployment across different LLM providers.To address these limitations, the proposed Multi-Agent Orchestrator introduces a modular and stateful architecture for repository-level issue resolution. A centralized LangGraph-based orchestrator coordinates specialized agents responsible for issue analysis, repository understanding, solution planning, code generation, testing, code review, and pull request preparation. Retrieval-Augmented Generation (RAG) is integrated to selectively retrieve only the most relevant repository context, reducing computational overhead while improving response quality.The architecture is further designed to support multiple LLM providers through a configurable interface, enabling flexible and cost-effective deployment. The project focuses on automating repository-level GitHub issue resolution while preserving human oversight for complex architectural decisions and does not aim to replace software developers or fully automate every software engineering activity. The proposed system will be evaluated based on its ability to accurately retrieve relevant repository context, generate meaningful and compilable code modifications, automate major stages of the issue-resolution workflow, and reduce manual developer effort, thereby providing a scalable, modular, and practical framework for next-generation AI-assisted software engineering.

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References (Category specific)

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[2] Co-Evolve Lab,"MAGIS: LLM-Based Multi-Agent Framework for GitHub Issue Resolution," GitHub Repository.Available: https://github.com/co-evolve-lab/magis

[3] K. Koech, M. Adam, B. B. Jean Jacques, and J. Barros,"Phoenix: Safe GitHub Issue Resolution via Multi-Agent LLMs,"arXiv preprint arXiv:2606.20243, 2026.

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